

September 28, 2026

Editor-in-Chief
IEEE Transactions on Neural Networks and Learning Systems

Dear Editor-in-Chief,

I am pleased to submit the manuscript entitled “*DisCo-CFL: Calibrated, Fair and Explainable Clustered Federated Learning via Disattenuated Class-Conditional Gradient Signatures*” for consideration as a regular paper in the IEEE Transactions on Neural Networks and Learning Systems.

Clustered federated learning (CFL) trains one model per group of clients that share a data distribution. A recent survey of the field identifies several open problems: the clustering signal is high-dimensional and changes from round to round, methods tend to group clients by label distribution rather than by concept, the number of clusters is set through hyperparameters without a stable meaning, performance degrades under quantity skew, and grouping decisions are neither private nor explained. The manuscript addresses these problems with a single algorithm and its analysis.

The main contributions are:

- **A new CFL algorithm.** Each client uploads, once, sketched class-conditional gradient signatures computed on two random halves of its data. Spearman’s correction for attenuation, estimated from the two halves, yields a calibrated similarity that is invariant to label and quantity skew and equals one for clients that share a concept. A class-wise bottleneck linkage then determines the number of clusters from one threshold on this calibrated scale.
- **Guarantees beyond accuracy.** We prove safety (conflicting clients are never merged), maximality and exact recovery with a sample complexity that does not depend on cluster sizes, which gives fairness towards minority groups, and a minimax lower bound that matches it up to a factor $1/\Delta$. We also prove differential privacy with an explicit cost, the fidelity of the class-level explanations, and the Bayes optimality of prior-corrected cluster models. The deterministic core of all proofs (25 theorems) is machine-checked in Lean 4.
- **Transparency and accountability.** The server returns assignment certificates with uncertainty, an audit log, anomaly flags, class-level explanations of why clusters differ, and a test for newcomers with a novel concept.
- **Extensive evaluation.** 618 training runs on Fashion-MNIST, MNIST, SVHN and CIFAR-10 against nine baselines, with robustness sweeps, ablations and significance tests. DisCo-CFL recovers the concept groups without knowing their number and merges on average 0.1% of conflicting client pairs, against 8–84% for baselines that are given the true number of clusters. The limitations, including the weaker results on CIFAR-10 with a short warm-up, are reported explicitly.

We believe the manuscript fits the scope of the Transactions, as it combines a new learning algorithm for distributed neural network training with theoretical guarantees and a thorough empirical study. The main paper is 8 pages long. Full proofs, the Lean 4 formalisation and complete result tables are provided in a separate supplementary file. Code, raw results and the Lean project are publicly available at https://github.com/vinhqdang/cluster_federated_learning.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The author declares no conflict of interest. Thank you for considering this submission. I look forward to hearing from you.

Sincerely,

Quang-Vinh Dang
British University Vietnam